



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA0618 – DESIGN AND ANALYSIS OF ALGORITHM

EXPERIMENT NO: 07

Name: Avishek N

Reg No: 192511009

Program:

```
#include <stdio.h>

void quickSort(int a[], int low, int high)
{
    if (low < high)
    {
        int pivot = a[high];
        int i = low - 1, j, temp;
        for (j = low; j < high; j++)
        {
            if (a[j] < pivot)
            {
                i++;
                temp = a[i];
                a[i] = a[j];
                a[j] = temp;
            }
        }
        temp = a[i + 1];
        a[i + 1] = a[high];
```

```

        a[high] = temp;

    int p = i + 1;
    quickSort(a, low, p - 1);
    quickSort(a, p + 1, high);
}
}

int main()
{
    int a[] = {10, 7, 8, 9, 1, 5};
    int n = 6, i;
    quickSort(a, 0, n - 1);
    printf("[");
    for (i = 0; i < n; i++)
    {
        printf("%d", a[i]);
        if (i < n - 1)
            printf(", ");
    }
    printf("]");
    return 0;
}

```

Output:

main.c	Run ▶	Output
<pre>1 #include <stdio.h> 2 3 void quickSort(int a[], int low, int high) 4 { 5 if (low < high) 6 { 7 int pivot = a[high]; 8 int i = low - 1, j, temp; 9 10 for (j = low; j < high; j++) 11 { 12 if (a[j] < pivot) 13 { 14 i++; 15 temp = a[i]; 16 a[i] = a[j]; 17 a[j] = temp; 18 } 19 } 20 21 temp = a[i + 1]; 22 a[i + 1] = a[high]; 23 a[high] = temp; 24 25 int p = i + 1; 26 27 quickSort(a, low, p - 1); 28 quickSort(a, p + 1, high); 29 } 30 } 31 32 int main() 33 { 34 int a[] = {10, 7, 8, 9, 1, 5}; 35 int n = 6, i; 36 37 quickSort(a, 0, n - 1); 38 39 printf("["); 40 for (i = 0; i < n; i++) 41 { 42 printf("%d", a[i]); 43 if (i < n - 1) 44 printf(", "); 45 } 46 printf("]"); 47 48 return 0; 49 }</pre>		[1, 5, 7, 8, 9, 10]